

NORRKOPING BRAVALLA, Sweden

WMO: 025700

Lat: 58.617N

Lon: 16.117E

Elev: 108

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.5	5.9	-6.9	3.8	-0.3	-0.5	5.4	6.7	29.1	43.6	26.9	39.8	5.6	280	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	79.4	64.1	76.0	62.7	72.8	61.1	66.0	74.8	64.2	73.0	62.7	70.4	7.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	86.1	68.5	61.0	80.5	67.1	59.4	75.9	65.4	30.7	74.8	29.4	72.9	28.3	70.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	22.5	19.4	DB	-1.5	84.0	10.5	3.3	-9.0	86.4	-15.1	88.3	-21.0	90.2	-28.6	92.6	(4)
(5)			WB	-2.2	69.5	10.4	2.7	-9.6	71.5	-15.7	73.1	-21.6	74.6	-29.1	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	28.0	27.4	33.8	40.6	51.9	58.1	62.3	60.3	52.2	44.9	36.1	30.6		
	DBStd	14.17	11.12	10.48	7.18	6.23	5.97	5.72	4.39	4.86	4.80	5.88	6.74	7.96		
	HDD50	3384	683	632	504	288	49	5	0	0	31	172	419	601		
	HDD65	7734	1148	1052	968	732	406	219	108	161	384	622	868	1066		
	CDD50	1172	0	0	1	6	108	246	380	319	97	15	0	0		
	CDD65	48	0	0	0	0	0	11	23	14	0	0	0	0		
	CDH74	496	0	0	0	0	24	104	234	133	1	0	0	0		
	CDH80	65	0	0	0	0	0	9	33	23	0	0	0	0		
(14) Wind	WSAvg	8.9	10.7	9.4	9.3	8.6	9.0	7.6	7.5	7.7	9.0	9.5	9.3	9.4		
(15) Precipitation	PrecAvg	24.1	1.7	1.4	1.2	1.4	1.7	2.7	2.9	2.8	2.3	2.2	2.0	1.9		
	PrecMax	31.3	3.4	2.4	2.2	3.5	3.5	4.9	5.6	6.9	5.1	5.5	4.6	3.5		
	PrecMin	17.2	0.3	0.6	0.3	0.1	0.3	0.5	0.2	0.4	0.4	0.1	0.4	0.5		
	PrecStd	2.9	0.8	0.5	0.6	0.9	0.8	1.1	1.3	1.3	1.3	1.1	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.6	51.5	55.6	66.0	78.3	81.2	83.5	83.3	72.2	62.2	52.2	48.6		
		MCWB	45.4	46.8	48.0	52.5	61.3	64.3	66.3	65.1	63.0	57.4	49.1	45.7		
	2%	DB	44.8	47.1	49.5	59.6	73.4	76.9	80.1	77.7	65.5	57.4	50.0	45.4		
		MCWB	42.4	44.1	43.7	49.0	58.4	62.6	64.7	64.1	57.5	53.9	47.1	43.2		
	5%	DB	42.5	43.5	46.0	54.9	68.7	73.5	76.3	73.0	62.6	55.2	47.4	42.6		
		MCWB	40.1	41.1	41.8	46.6	55.4	60.3	63.5	62.1	55.9	51.6	44.9	40.6		
	10%	DB	40.3	39.9	43.0	51.7	64.3	69.6	72.8	69.7	60.3	53.3	45.2	40.2		
		MCWB	38.1	38.2	39.9	45.0	53.3	58.3	62.0	60.6	54.8	50.0	43.2	38.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	47.4	48.3	53.5	63.0	66.6	69.4	68.3	63.7	58.2	50.1	45.7		
		MCDB	47.9	50.8	54.2	64.3	75.0	76.5	79.5	75.7	69.4	60.2	52.2	47.7		
	2%	WB	42.6	44.1	44.8	49.9	59.3	64.4	66.4	65.4	60.1	54.8	47.3	43.5		
		MCDB	44.4	46.6	47.9	57.1	71.2	73.8	76.0	73.6	63.4	56.9	49.4	45.4		
	5%	WB	40.4	41.2	42.3	47.7	56.6	62.2	64.5	63.5	57.5	52.6	45.3	40.9		
		MCDB	42.3	43.2	44.9	54.3	66.1	70.4	73.2	71.6	61.2	54.6	47.1	42.4		
	10%	WB	38.3	38.3	40.1	45.5	54.4	60.1	62.9	61.8	55.5	50.5	43.5	38.8		
		MCDB	40.1	39.8	42.7	50.5	62.7	68.0	70.7	68.4	58.9	52.6	45.0	40.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.8	10.6	14.4	17.9	16.2	16.8	15.5	13.4	10.7	7.9	8.6		
		MCDBR	8.4	9.5	14.8	21.3	26.1	22.9	24.2	21.4	16.3	12.3	9.3	9.8		
		MCWBR	7.8	8.0	10.0	13.0	14.2	10.6	11.4	10.1	10.0	9.2	8.6	9.4		
	5% WB	MCDBR	8.3	9.3	12.5	19.0	22.4	20.1	20.5	18.5	14.2	11.3	8.6	9.7		
		MCWBR	8.0	8.2	9.4	12.2	13.2	10.3	10.8	10.0	9.7	9.2	8.3	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.321	0.340	0.378	0.374	0.363	0.390	0.381	0.348	0.338	0.310	0.270		
	taud		2.241	2.295	2.350	2.308	2.385	2.441	2.386	2.408	2.484	2.419	2.289	2.175		
	Ebn at Noon		173	222	254	262	272	277	266	259	251	215	160	134		
	Edh at Noon		14	22	28	35	34	33	34	31	25	20	14	11		
(44) All-Sky Solar Radiation	RadAvg		107	296	773	1262	1659	1840	1669	1327	891	400	138	74		
	RadStd		15	50	81	130	137	86	167	128	111	62	29	9		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (55 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	-230	-286	N/A	N/A	(47)

Nomenclature: See separate page